PROJECT UNDERSTANDING DOCUMENT

Enterprise Healthcare Agentic AI Copilot

Contents:

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1.Introduction:

Artificial Intelligence is transforming healthcare by enabling intelligent retrieval, analysis, and decision support from large volumes of structured and unstructured healthcare data. Modern healthcare enterprises generate enormous amounts of information including patient records, prescriptions, insurance documents, hospital policies, clinical reports, and operational workflows.

MediAssist AI is designed as an Enterprise Healthcare Agentic AI Copilot capable of integrating multiple healthcare data sources into a unified intelligent platform. The system leverages Generative AI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Multimodal AI, PostgreSQL integration through MCP, and cloud deployment to provide accurate and context-aware healthcare assistance.

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2. Project Overview:

MediAssist AI is an enterprise-grade Healthcare AI Copilot designed to help hospitals and healthcare organizations access information from multiple data sources through a single intelligent interface. The system combines Generative AI, Retrieval-Augmented Generation (RAG), Multi-Agent AI, Multimodal AI, and Model Context Protocol (MCP) integration to provide accurate and context-aware responses.

Healthcare organizations maintain large amounts of structured data such as patient records, appointments, prescriptions, billing details, and insurance information, along with unstructured documents such as SOPs, compliance guidelines, discharge summaries, and medical reports. MediAssist AI acts as a unified assistant capable of understanding, retrieving, and reasoning over these diverse data sources.

The platform supports conversational interactions, document understanding, image analysis, enterprise database access, and workflow automation, making healthcare operations faster and more efficient.

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3. Problem Statement

Modern hospitals generate huge amounts of data every day. This information is often distributed across different systems and departments, making it difficult for healthcare professionals to retrieve relevant information quickly.

Some common challenges include:

- Information stored in isolated systems.
- Time-consuming manual searches.
- Delayed access to patient history.
- Difficulty analyzing large volumes of healthcare documents.
- Lack of integration between databases and AI systems.
- High operational overhead for repetitive tasks.

As healthcare organizations grow across multiple locations, these problems become more severe. Therefore, there is a need for an intelligent AI-based solution capable of integrating multiple data sources and providing accurate responses through a unified interface.

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4. Project Objectives:

The primary objectives of the project are:

Enterprise Information Retrieval

Develop a system capable of retrieving information from healthcare documents and databases using natural language queries.

Agentic AI Workflow

Implement multiple AI agents that collaborate to solve complex healthcare tasks through task decomposition and orchestration.

Multimodal Understanding

Enable the system to understand uploaded prescriptions, reports, and healthcare images using vision-language models.

Database Integration

Connect enterprise healthcare databases using MCP architecture to allow secure AI-driven access to structured data.

Workflow Automation

Automate repetitive healthcare processes such as patient summary generation, appointment lookup, and report analysis.

Enterprise Deployment

Deploy the complete solution using Docker containers and Azure Cloud infrastructure.

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5. Scope of the Project

The project scope includes:

Structured Data Retrieval

- Patient records
- Doctor information
- Billing details
- Lab results
- Appointments

Unstructured Data Retrieval

- Hospital SOPs
- Insurance documents
- Compliance policies
- Healthcare guidelines
- Operational manuals

Image Processing

- Prescription understanding
- Medical report interpretation
- Lab report extraction

Analytics

- Admission trends
- Department performance analysis
- Insurance claim statistics
- Branch-level reporting

The project focuses on information retrieval and decision support rather than direct patient treatment or clinical decision-making.

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6. Existing System Analysis

Current healthcare systems suffer from:

• Data fragmentation

• Manual search processes

• Slow response times

• Limited interoperability

• Poor integration with AI technologies

Healthcare professionals often spend significant time searching for information across multiple systems.

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7. Proposed System

The proposed solution introduces a unified AI-powered healthcare assistant capable of:

• Understanding natural language queries

• Accessing enterprise databases

• Retrieving healthcare documents

• Understanding uploaded images

• Coordinating multiple AI agents

• Generating grounded recommendations

The solution significantly improves accessibility, efficiency, and decision support.

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8. Business Requirements:

The healthcare organization requires:

• Centralized information retrieval

• Secure database access

• Automated workflows

• Multi-branch analytics

• Compliance support

• AI-powered assistance

• Scalable cloud deployment

9.System Architecture:

Architecture Layers:

1. User Interface Layer (Streamlit)
2. Backend Layer (FastAPI)
3. Orchestration Layer (LangGraph)
4. Agent Layer
5. Retrieval Layer
6. Database Layer
7. Evaluation Layer

Workflow:

User → Streamlit → FastAPI → LangGraph → Agents → Database/RAG → Evaluation → Response

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10. Technology Stack:

Frontend:
• Streamlit

Backend:
• FastAPI

LLM Framework:
• LangChain

Agent Framework:
• LangGraph

LLM:
• OpenAI / Groq

Embeddings:
• Hugging Face

Vector Databases:
• FAISS
• ChromaDB

Database:
• PostgreSQL

Cloud:
• Azure

Containerization:
• Docker

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11. Project Modules

Module 1 – Data Ingestion Pipeline

Module 2 – Enterprise RAG Pipeline

Module 3 – Multi-Agent System

Module 4 – MCP Server

Module 5 – Multimodal AI

Module 6 – Evaluation & Observability

Module 7 – Security & Guardrails

Module 8 – Dockerization

Module 9 – Azure Deployment

12. Multi-Agent Architecture

- Planner Agent

  Responsible for task decomposition and workflow planning.

- Retriever Agent

  Responsible for enterprise document retrieval.

- MCP Agent

  Responsible for structured database access.

- Multimodal Agent

  Responsible for image understanding and extraction.

- Reasoning Agent

  Responsible for generating final responses.

- Evaluation Agent

  Responsible for validating output quality.

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13. MCP Integration

Model Context Protocol (MCP) acts as a secure interface between AI agents and PostgreSQL databases.

Example Tools:

get_patient_history()

get_appointments()

get_lab_results()

get_prescriptions()

get_billing_summary()

This architecture improves security and scalability.

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14. Enterprise RAG Pipeline

The RAG pipeline consists of:

Document Ingestion

Text Chunking

Embedding Generation

Vector Indexing

Semantic Retrieval

Context Grounding

The retrieved context is supplied to the LLM for accurate response generation.

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15. Multimodal AI Module

This module processes:

• Prescriptions

• Medical Reports

• Lab Results

Capabilities:

• OCR

• Information Extraction

• Report Understanding

• Medical Summarization

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16. Database Design

Major Tables:

Patients

Admissions

Appointments

Doctors

Departments

Billing

Lab Results

Prescriptions

Insurance Policies

Hospital Branches

Total Records: Approximately 1.1 Million

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17. Functional Requirements

• Upload documents

• Upload images

• Ask healthcare queries

• Retrieve patient data

• Generate summaries

• Analyze reports

• Access enterprise records

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18. Non-Functional Requirements

• Scalability

• Security

• Reliability

• Availability

• Performance

• Logging

• Monitoring

• Error Handling

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19. User Personas

Doctor

Hospital Administrator

Insurance Team

Clinical Operations Team

Front Desk Staff

20. Use Cases

Operational Queries

Analytical Queries

Multimodal Queries

Multi-Agent Queries

Enterprise Reporting

21. Workflow Execution

User Query

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Planner Agent

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Retriever + MCP + Multimodal Agents

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Reasoning Agent

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Evaluation Agent

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Final Response

22. Evaluation Framework

LLM-as-a-Judge

Faithfulness Checking

Hallucination Detection

Grounding Validation

Response Relevance

Confidence Scoring

23. Security and Guardrails

Prompt Injection Protection

Sensitive Data Protection

Unsafe Recommendation Detection

Tool Misuse Prevention

Hallucination Prevention

24. Dockerization and Deployment

Dockerfile Creation

Docker Compose Setup

Containerized Services

Azure Container Apps Deployment

25. Expected Outcomes

- Faster healthcare information retrieval
- Reduced operational effort
- Improved patient support
- Better enterprise insights
- Secure AI-driven workflows

26. Future Enhancements

- Voice Assistant
- Mobile Application
- EHR Integration
- Predictive Analytics
- Real-Time Monitoring
- Multi-Language Support

27. Conclusion

MediAssist AI represents a modern enterprise healthcare platform that combines Generative AI, RAG, Multi-Agent Systems, MCP Integration, Multimodal AI, and Cloud Deployment. The project demonstrates how advanced AI technologies can improve healthcare operations, information retrieval, and enterprise decision-making.